

Laxmi Sai Maneesh Reddy Jupalle

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SUMMARY

Computer Science graduate student at the University of Illinois at Chicago with 5+ years of experience in distributed systems, full-stack development, and enterprise software support. Hands-on experience building GenAI systems including RAG pipelines, LLM-driven applications, and vector-database-backed retrieval workflows. Strong background in debugging production systems, working across APIs and cloud platforms, and designing scalable, distributed architectures.

EDUCATION

University of Illinois at Chicago (UIC) <i>Master of Science in Computer Science</i> (GPA: 3.80) Coursework: Distributed Systems, Artificial Intelligence, Machine Learning, Data Science, Networking, DBMS, DevOps, GenAI.	Aug 2024 – May 2026 Chicago, IL
Jain (Deemed to be University) <i>B.Tech in Computer Science and Technology in Artificial Intelligence</i> (GPA: 3.72) Coursework: Machine Learning, Deep Learning, NLP, HCI, Data Structures, DSA, Distributed Systems, DBMS, Data Science, AI	Jul 2018 – Jul 2022 Bengaluru, KA

RESEARCH EXPERIENCE

AI-Driven Academic Research Assistant <i>Independent Research Project – University of Illinois at Chicago</i>	Dec 2024 – Jan 2025
- Investigated multi-LLM architectures for automated academic literature analysis, designing a system capable of extractive and abstractive summarization, citation-aware retrieval, and contextual question answering over research corpora. - Implemented and evaluated a full-stack research prototype (React, TypeScript, Node.js, Supabase) with JWT-based authentication and row-level security, enabling real-time iterative exploration of uploaded papers through conversational AI interfaces.	

WORK EXPERIENCE

University of Illinois at Chicago <i>Graduate Assistant – GenAI Team</i>	Chicago, IL Jan 2026 – May 2026
Architecting an end-to-end RAG-based clinical AI system using Python, Azure AI Foundry, and open-source LLMs (LLaMA, Mistral) with vector-database retrieval to automate medical justification reports for assistive technology equipment. Developing secure, HIPAA-compliant data workflows to process, anonymize, and validate patient records from protected file servers, grounding AI-generated clinical documentation in verified inventory and patient datasets.	
VMware by Broadcom <i>Software Engineer, Workspace ONE</i>	Bengaluru, KA Aug 2022 – Jul 2024
Resolved 250+ production-impacting cases monthly involving REST APIs, certificates, XML payloads, and profile-based configurations across Apple-VMware MDM integrations, maintaining strict SLA and uptime requirements. Performed deep log aggregation and configuration audits to isolate failures in distributed enterprise environments, reducing mean time to resolution across P1/P2 severity incidents. Authored reproducible Jira cases with detailed logs and environment metadata, streamlining handoff to engineering teams and reducing escalation cycles by 40%. Created comprehensive knowledge base articles and internal runbooks for recurring enterprise issues, enabling faster onboarding of new engineers and reducing repeat escalation volume across the team.	

TECHNICAL SKILLS

Programming Languages : C++, Python, Java, JavaScript, TypeScript, Rust, SQL
Frameworks & Libraries : React, Django, TensorFlow, PyTorch, Scikit-Learn, LangChain, LangGraph, Hugging Face
AI & Data : RAG, Generative AI, Data Analysis, PowerBI, Tableau
Cloud & DevOps : AWS, Azure, Docker, Jenkins, Git, Linux
Languages : English, Telugu, Kannada, Hindi, Japanese

PROJECTS

Golden Bridge: Ambulance Pre-Arrival System (<i>1st Place, Microsoft-Sponsored Hack with Chicago</i>)	Dec 2025
Architected a multi-agent orchestration system using LangChain and GPT-4 for intelligent triage automation with coordinated diagnostic, severity scoring, and treatment agents. Engineered real-time streaming using Pathway, Kafka, FastAPI, and WebSockets for sub-second latency EMS data transmission, deployed on Azure using Docker.	
Insurance Document Intelligence Platform	Jan 2025 – May 2025
Built a web-based LLM-powered document analysis system using GPT-Nano/Gemini Flash APIs with vector and graph search (Qdrant, Neo4j) to extract, compare, and simplify insurance documents. Designed a scalable modular architecture with Extract, Query, Analyze, and Compare pipelines to enable structured decision-making through integrated LLM workflows.	

PUBLICATIONS

Diabetic Retinopathy Detection using ML	Jan 2022 – Jun 2022
- Built a CNN-based diabetic retinopathy detection system; received International Best Researcher recognition. - www.ijariit.com/manuscript/diabetic-retinopathy-detection-using-machine-learning/	